

Optimization Lecture 6+7

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Lagrangian and dual function

Lagrangian dual problem

KKT condition

Sensitivity analysis

Problem reformulations

Theorems of alternatives

Recap

- ▶ conjugate of $f(x)$
- ▶ first order condition for a differentiable convex function
- ▶ Jensen inequality
- ▶ implicit constraints, explicit constraints
- ▶ first order sufficient and necessary optimality condition for convex optimization problem.

Lagrangian and dual function

Lagrangian

- ▶ standard form problem (not necessarily convex)

$$\begin{array}{ll}\text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & h_i(x) = 0, \quad i = 1, \dots, p\end{array}$$

variable $x \in \mathbf{R}^n$, domain $\mathcal{D}$, optimal value p^*

- ▶ **Lagrangian:** $L : \mathbf{R}^n \times \mathbf{R}^m \times \mathbf{R}^p \rightarrow \mathbf{R}$, with $\text{dom } L = \mathcal{D} \times \mathbf{R}^m \times \mathbf{R}^p$,

$$L(x, \lambda, v) = f_0(x) + \sum_{i=1}^m \lambda_i f_i(x) + \sum_{i=1}^p v_i h_i(x)$$

- ▶ weighted sum of objective and constraint functions
- ▶ λ_i is Lagrange multiplier associated with $f_i(x) \leq 0$
- ▶ v_i is Lagrange multiplier associated with $h_i(x) = 0$

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Least-norm solution of linear equations

$$\begin{array}{ll}\text{minimize} & x^T x \\ \text{subject to} & Ax = b\end{array}$$

- ▶ Lagrangian is $L(x, v) = x^T x + v^T (Ax - b)$
- ▶ to minimize L over x , set gradient equal to zero:

$$\nabla_x L(x, v) = 2x + A^T v = 0 \implies x = -(1/2)A^T v$$

- ▶ plug x into L to obtain

$$g(v) = L(-(1/2)A^T v, v) = -\frac{1}{4}v^T A A^T v - b^T v$$

- ▶ lower bound property: $p^* \geq -(1/4)v^T A A^T v - b^T v$ for all v

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Lagrange dual function

Lagrange dual function: $g : \mathbf{R}^m \times \mathbf{R}^p \rightarrow \mathbf{R}$,

$$g(\lambda, v) = \inf_{x \in \mathcal{D}} L(x, \lambda, v) = \inf_{x \in \mathcal{D}} \left(f_0(x) + \sum_{i=1}^m \lambda_i f_i(x) + \sum_{i=1}^p v_i h_i(x) \right)$$

- ▶ g is concave, can be $-\infty$ for some λ, v
- ▶ lower bound property: if $\lambda \geq 0$, then $g(\lambda, v) \leq p^*$
- ▶ proof: if $\tilde{x}$ is feasible and $\lambda \geq 0$, then

$$f_0(\tilde{x}) \geq L(\tilde{x}, \lambda, v) \geq \inf_{x \in \mathcal{D}} L(x, \lambda, v) = g(\lambda, v)$$

minimizing over all feasible $\tilde{x}$ gives $p^* \geq g(\lambda, v)$

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Standard form LP

$$\begin{array}{ll}\text{minimize} & c^T x \\ \text{subject to} & Ax = b, \quad x \geq 0\end{array}$$

- ▶ Lagrangian is

$$L(x, \lambda, v) = c^T x + v^T (Ax - b) - \lambda^T x = -b^T v + (c + A^T v - \lambda)^T x$$

- ▶ L is affine in x , so

$$g(\lambda, v) = \inf_x L(x, \lambda, v) = \begin{cases} -b^T v & A^T v - \lambda + c = 0 \\ -\infty & \text{otherwise} \end{cases}$$

- ▶ g is linear on affine domain $\{(\lambda, v) \mid A^T v - \lambda + c = 0\}$, hence concave
- ▶ lower bound property: $p^* \geq -b^T v$ if $A^T v + c \geq 0$

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Equality constrained norm minimization

$$\begin{array}{ll}\text{minimize} & \|x\| \\ \text{subject to} & Ax = b\end{array}$$

- ▶ dual function is

$$g(v) = \inf_x (\|x\| - v^T Ax + b^T v) = \begin{cases} b^T v & \|A^T v\|_* \leq 1 \\ -\infty & \text{otherwise} \end{cases}$$

where $\|v\|_* = \sup_{\|u\| \leq 1} u^T v$ is dual norm of $\|\cdot\|$

- ▶ lower bound property: $p^* \geq b^T v$ if $\|A^T v\|_* \leq 1$

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Lagrange dual and conjugate function

$$\begin{array}{ll}\text{minimize} & f_0(x) \\ \text{subject to} & Ax \leq b, \quad Cx = d\end{array}$$

- ▶ dual function

$$\begin{aligned} g(\lambda, v) &= \inf_{x \in \text{dom } f_0} (f_0(x) + (A^T \lambda + C^T v)^T x - b^T \lambda - d^T v) \\ &= -f_0^*(-A^T \lambda - C^T v) - b^T \lambda - d^T v \end{aligned}$$

where $f^*(y) = \sup_{x \in \text{dom } f} [y^T x - f(x)]$ is the conjugate of f_0

- ▶ simplifies derivation of dual if conjugate of f_0 is known
- ▶ example: entropy maximization

$$f_0(x) = \sum_{i=1}^n x_i \log x_i, \quad f_0^*(y) = \sum_{i=1}^n e^{y_i - 1}$$

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The Lagrange dual problem

(Lagrange) dual problem

$$\begin{array}{ll}\text{maximize} & g(\lambda, v) \\ \text{subject to} & \lambda \geq 0\end{array}$$

- ▶ finds best lower bound on p^* , obtained from Lagrange dual function
- ▶ a convex optimization problem, even if original primal problem is not
- ▶ dual optimal value denoted d^*
- ▶ λ, v are dual feasible if $\lambda \geq 0, (\lambda, v) \in \text{dom } g$
- ▶ often simplified by making implicit constraint $(\lambda, v) \in \text{dom } g$ explicit

Lagrangian dual problem

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Example: standard form LP

- ▶ primal standard form LP:

$$\begin{array}{ll}\text{minimize} & c^T x \\ \text{subject to} & Ax = b \\ & x \geq 0\end{array}$$

- ▶ dual problem is

$$\begin{array}{ll}\text{maximize} & g(\lambda, v) \\ \text{subject to} & \lambda \geq 0\end{array}$$

with $g(\lambda, v) = -b^T v$ if $A^T v - \lambda + c = 0$, $-\infty$ otherwise

- ▶ make implicit constraint explicit, and eliminate λ to obtain (transformed) dual problem

$$\begin{array}{ll}\text{maximize} & -b^T v \\ \text{subject to} & A^T v + c \geq 0\end{array}$$

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Weak and strong duality

weak duality: $d^* \leq p^*$

- ▶ always holds (for convex and nonconvex problems)
- ▶ can be used to find nontrivial lower bounds for difficult problems.

strong duality: $d^* = p^*$

- ▶ does not hold in general
- ▶ (usually) holds for convex problems
- ▶ conditions that guarantee strong duality in convex problems are called constraint qualifications

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Slater's constraint qualification

strong duality holds for a convex problem

$$\begin{array}{ll}\text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & Ax = b\end{array}$$

if it is strictly feasible, i.e., there is an $x \in \text{int } \mathcal{D}$ with $f_i(x) < 0, i = 1, \dots, m, Ax = b$

- ▶ also guarantees that the dual optimum is attained (if $p^* > -\infty$)
- ▶ can be sharpened: e.g.,
 - ▶ can replace $\text{int } \mathcal{D}$ with $\text{relint } \mathcal{D}$ (interior relative to affine hull)
 - ▶ linear inequalities do not need to hold with strict inequality
- ▶ there are many other types of constraint qualifications

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Inequality form LP

primal Problem

$$\begin{array}{ll}\text{minimize} & c^T x \\ \text{subject to} & Ax \leq b\end{array}$$

dual function

$$g(\lambda) = \inf_x \left((c + A^T \lambda)^T x - b^T \lambda \right) = \begin{cases} -b^T \lambda & A^T \lambda + c = 0 \\ -\infty & \text{otherwise} \end{cases}$$

dual problem

$$\begin{array}{ll}\text{maximize} & -b^T \lambda \\ \text{subject to} & A^T \lambda + c = 0, \quad \lambda \geq 0\end{array}$$

- ▶ $p^* = d^*$ except when primal and dual are both infeasible (See books on LP)

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Quadratic program

primal problem (assume $P \in \mathbf{S}_{++}^n$)

$$\begin{array}{ll} \text{minimize} & x^T P x \\ \text{subject to} & Ax \leq b \end{array}$$

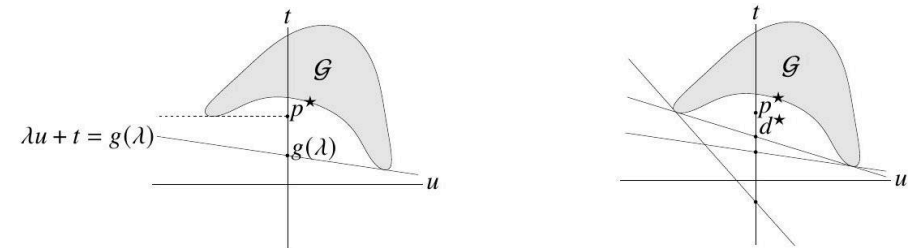
dual function

$$g(\lambda) = \inf_x [x^T P x + \lambda^T (A x - b)] = -\frac{1}{4} \lambda^T A P^{-1} A^T \lambda - b^T \lambda$$

dual problem

$$\begin{array}{ll} \text{maximize} & -(1/4)\lambda^T A P^{-1} A^T \lambda - b^T \lambda \\ \text{subject to} & \lambda \geq 0 \end{array}$$

- ▶ from the sharpened Slater's condition: $p^* = d^*$ if the primal problem is feasible
- ▶ in fact, $p^* = d^*$ always



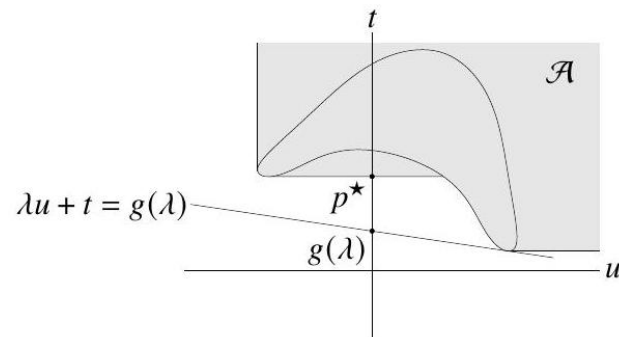
- ▶ for simplicity, consider problem with one constraint $f_1(x) \leq 0$
- ▶ $\mathcal{G} = \{(f_1(x), f_0(x)) \mid x \in \mathcal{D}\}$ is set of achievable (constraint, objective) values
- ▶ interpretation of dual function: $g(\lambda) = \inf_{(u,t) \in \mathcal{G}} (t + \lambda u)$
- ▶ $\lambda u + t = g(\lambda)$ is (non-vertical) supporting hyperplane to $\mathcal{G}$
- ▶ hyperplane intersects t -axis at $t = g(\lambda)$

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Epigraph variation

- ▶ same with $\mathcal{G}$ replaced with $\mathcal{A} = \{(u, t) \mid f_1(x) \leq u, f_0(x) \leq t \text{ for some } x \in \mathcal{D}\}$



KKT condition

- ▶ strong duality holds if there is a non-vertical supporting hyperplane to $\mathcal{A}$ at $(0, p^*)$
- ▶ for convex problem, $\mathcal{A}$ is convex, hence has supporting hyperplane at $(0, p^*)$
- ▶ Slater's condition: if there exist $(\tilde{u}, \tilde{t}) \in \mathcal{A}$ with $\tilde{u} < 0$, then supporting hyperplane at $(0, p^*)$ must be non-vertical

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Complementary slackness

- ▶ assume strong duality holds, x^* is primal optimal, (λ^*, v^*) is dual optimal

$$\begin{aligned} f_0(x^*) &= g(\lambda^*, \nu^*) = \inf_x \left(f_0(x) + \sum_{i=1}^m \lambda_i^* f_i(x) + \sum_{i=1}^p \nu_i^* h_i(x) \right) \\ &\leq f_0(x^*) + \sum_{i=1}^m \lambda_i^* f_i(x^*) + \sum_{i=1}^p \nu_i^* h_i(x^*) \\ &\leq f_0(x^*) \end{aligned}$$

- ▶ hence, the two inequalities hold with equality
- ▶ x^* minimizes $L(x, \lambda^*, v^*)$
- ▶ $\lambda_i^* f_i(x^*) = 0$ for $i = 1, \dots, m$ (known as complementary slackness):

$$\lambda_i^* > 0 \implies f_i(x^*) = 0, \quad f_i(x^*) < 0 \implies \lambda_i^* = 0$$

Karush-Kuhn-Tucker (KKT) conditions

the KKT conditions (for a problem with differentiable f_i, h_i) are

1. primal constraints: $f_i(x) \leq 0, i = 1, \dots, m, h_i(x) = 0, i = 1, \dots, p$
2. dual constraints: $\lambda \geq 0$
3. complementary slackness: $\lambda_i f_i(x) = 0, i = 1, \dots, m$
4. gradient of Lagrangian with respect to x vanishes:

$$\nabla f_0(x) + \sum_{i=1}^m \lambda_i \nabla f_i(x) + \sum_{i=1}^p \nu_i \nabla h_i(x) = 0$$

if strong duality holds and x, λ, v are optimal, they satisfy the KKT conditions (why?)

KKT conditions for convex problem

if $\tilde{x}, \tilde{\lambda}, \tilde{v}$ satisfy KKT for a convex problem, then they are optimal:

- ▶ from complementary slackness: $f_0(\tilde{x}) = L(\tilde{x}, \tilde{\lambda}, \tilde{\nu})$
- ▶ from 4th condition (and convexity): $g(\tilde{\lambda}, \tilde{\nu}) = L(\tilde{x}, \tilde{\lambda}, \tilde{\nu})$

hence, $f_0(\tilde{x}) = g(\tilde{\lambda}, \tilde{v})$

if Slater's condition is satisfied, then:

x is optimal if and only if there exist λ, ν that satisfy KKT conditions

- ▶ recall that Slater implies strong duality, and dual optimum is attained
- ▶ generalizes optimality condition $\nabla f_0(x) = 0$ for unconstrained problem

Sensitivity analysis

(unperturbed) optimization problem and its dual

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & h_j(x) = 0, \quad i = 1, \dots, p \end{array} \qquad \begin{array}{ll} \text{maximize} & g(\lambda, \nu) \\ \text{subject to} & \lambda \geq 0 \end{array}$$

perturbed problem and its dual

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq u_i, \quad i = 1, \dots, m \\ & h_j(x) = v_j, \quad j = 1, \dots, p \end{array} \quad \begin{array}{ll} \text{maximize} & g(\lambda, v) - u^T \lambda - v^T v \\ \text{subject to} & \lambda \geq 0 \end{array}$$

- ▶ x is primal variable; u, v are parameters
- ▶ $p^*(u, v)$ is optimal value as a function of u, v
- ▶ $p^*(0, 0)$ is optimal value of unperturbed problem

- ▶ assume strong duality holds for unperturbed problem, with λ^*, v^* dual optimal
- ▶ apply weak duality to perturbed problem:

$$p^*(u, v) \geq g(\lambda^*, v^*) - u^T \lambda^* - v^T v^* = p^*(0, 0) - u^T \lambda^* - v^T v^*$$

- ▶ implications
 - ▶ if λ_i^* large: p^* increases greatly if we tighten constraint i ($u_i < 0$)
 - ▶ if λ_i^* small: p^* does not decrease much if we loosen constraint i ($u_i > 0$)
 - ▶ if v_i^* large and positive: p^* increases greatly if we take $v_i < 0$
 - ▶ if v_i^* large and negative: p^* increases greatly if we take $v_i > 0$
 - ▶ if v_i^* small and positive: p^* does not decrease much if we take $v_i > 0$
 - ▶ if v_i^* small and negative: p^* does not decrease much if we take $v_i < 0$

Local sensitivity via duality

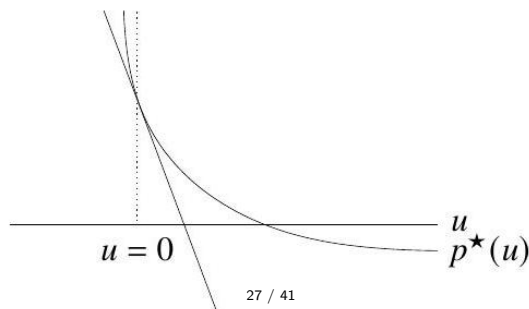
if (in addition) $p^*(u, v)$ is differentiable at $(0, 0)$, then

$$\lambda_i^* = -\frac{\partial p^*(0,0)}{\partial u_i}, \quad v_i^* = -\frac{\partial p^*(0,0)}{\partial v_i}$$

proof (for λ_i^*): from global sensitivity result,

$$\begin{aligned}\frac{\partial p^*(0,0)}{\partial u_i} &= \lim_{t \searrow 0} \frac{p^*(te_i, 0) - p^*(0, 0)}{t} \geq -\lambda_i^* \\ \frac{\partial p^*(0,0)}{\partial u_i} &= \lim_{t \nearrow 0} \frac{p^*(te_i, 0) - p^*(0, 0)}{t} \leq -\lambda_i^*\end{aligned}$$

hence, equality $p^*(u)$ for a problem with one (inequality) constraint:



Problem reformulations

Duality and problem reformulations

- ▶ equivalent formulations of a problem can lead to very different duals
- ▶ reformulating primal problem can be useful when dual is difficult to derive, or uninteresting

common reformulations

- ▶ introduce new variables and equality constraints
- ▶ make explicit constraints implicit or vice-versa
- ▶ transform objective or constraint functions, e.g., replace $f_0(x)$ by $\phi(f_0(x))$ with ϕ convex, increasing

Introducing new variables and equality constraints

- ▶ unconstrained problem: minimize $f_0(Ax + b)$
- ▶ dual function is constant: $g = \inf_x L(x) = \inf_x f_0(Ax + b) = p^*$
- ▶ we have strong duality, but dual is quite useless
- ▶ introduce new variable y and equality constraints $y = Ax + b$

$$\begin{array}{ll} \text{minimize} & f_0(y) \\ \text{subject to} & Ax + b - y = 0 \end{array}$$

- ▶ dual of reformulated problem is

$$\begin{array}{ll} \text{maximize} & b^T v - f_0^*(v) \\ \text{subject to} & A^T v = 0 \end{array}$$

- ▶ a nontrivial, useful dual (assuming the conjugate f_0^* is easy to express)

Example: Norm approximation

- ▶ minimize $\|Ax - b\|$
- ▶ reformulate as minimize $\|y\|$ subject to $y = Ax - b$
- ▶ recall conjugate of general norm:

$$\|z\|^* = \begin{cases} 0 & \|z\|_* \leq 1 \\ \infty & \text{otherwise} \end{cases}$$

- ▶ dual of (reformulated) norm approximation problem:

$$\begin{array}{ll} \text{maximize} & b^T v \\ \text{subject to} & A^T v = 0, \quad \|v\|_* \leq 1 \end{array}$$

Theorems of alternatives

Theorems of alternatives

- ▶ consider two systems of inequality and equality constraints
- ▶ called weak alternatives if no more than one system is feasible
- ▶ called strong alternatives if exactly one of them is feasible
- ▶ examples: for any $a \in \mathbf{R}$, with variable $x \in \mathbf{R}$,
 - ▶ $x > a, x < b$ and $x > b, x < a$ are weak alternatives
- ▶ a theorem of alternatives states that two inequality systems are (weak or strong) alternatives
- ▶ can be considered the extension of duality to feasibility problems

System A is called a strong alternative to System B iff exactly one is feasible.

System A is called a weak alternative to System B if that system A is feasible implies B is infeasible.

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Duality for feasibility problems

- ▶ dual function of feasibility problem is
$$g(\lambda, v) = \inf_x \left(\sum_{i=1}^m \lambda_i f_i(x) + \sum_{i=1}^p v_i h_i(x) \right)$$
- ▶ for $\lambda \geq 0$, we have $g(\lambda, v) \leq p^*$
- ▶ it follows that feasibility of the inequality system

$$\lambda \geq 0, \quad g(\lambda, v) > 0$$

implies the original system is infeasible

- ▶ so this is a weak alternative to original system
- ▶ it is strong if f_i convex, h_i affine, and a constraint qualification holds
- ▶ g is positive homogeneous so we can write alternative system as

$$\lambda \geq 0, \quad g(\lambda, v) \geq 1$$

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Feasibility problems

- ▶ consider system of (not necessarily convex) inequalities and equalities

$$f_i(x) \leq 0, \quad i = 1, \dots, m, \quad h_i(x) = 0, \quad i = 1, \dots, p$$

- ▶ express as feasibility problem

$$\begin{array}{ll} \text{minimize} & 0 \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & h_i(x) = 0, \quad i = 1, \dots, p \end{array}$$

- ▶ if system is feasible, $p^* = 0$; if not, $p^* = \infty$

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Example: Nonnegative solution of linear equations

- ▶ consider system

$$Ax = b, \quad x \geq 0$$

$$\text{dual function is } g(\lambda, v) = \begin{cases} -v^T b & A^T v = \lambda \\ -\infty & \text{otherwise} \end{cases}$$

- ▶ can express strong alternative of $Ax = b, x \geq 0$ as

$$A^T v \geq 0, \quad v^T b \leq -1$$

(we can replace $v^T b \leq -1$ with $v^T b = -1$)

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Farkas' lemma

Farkas' lemma:

$$Ax \leq 0, c^T x < 0 \quad (1) \quad \text{and} \quad A^T y + c = 0, \quad y \geq 0 \quad (2)$$

are strong alternatives

Proof: Consider (primal) LP and its dual

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & Ax \leq 0 \end{array} \quad \begin{array}{ll} \text{maximize} & 0 \\ \text{subject to} & A^T y + c = 0, y \geq 0 \end{array}$$

- ▶ $p^* = 0$ or $-\infty$. And $d^* = 0$ or $-\infty$.
- ▶ If (1) is infeasible, then $p^* = 0$.
- ▶ If (1) is feasible, then $p^* = -\infty$.
- ▶ If (2) is feasible, then $d^* = 0$.
- ▶ If (2) is infeasible, then $d^* = -\infty$. Noting that $x = 0$ is feasible for the primal problem, Strong duality for LP holds (see slide on inequality form LP).
- ▶ (1) and (2) are strong alternatives

Investment arbitrage

- ▶ we invest x_j in each of n assets $1, \dots, n$ with prices $p_1, \dots, p_n$
- ▶ our initial cost is $p^T x$
- ▶ at the end of the investment period there are only m possible outcomes $i = 1, \dots, m$
- ▶ V_{ij} is the payoff or final value of asset j in outcome i
- ▶ first investment is risk-free (cash): $p_1 = 1$ and $V_{i1} = 1$ for all i
- ▶ arbitrage means there is x with $p^T x < 0, Vx \geq 0$
- ▶ arbitrage means we receive money up front, and our investment cannot lose
- ▶ standard assumption in economics: the prices are such that there is no arbitrage

another version

Let $A \in R^{m \times n}$ and $b \in R^m$. Either

- ▶ $Ax \leq b$ has a solution, or
- ▶ $yA = 0, y \geq 0, b^T y < 0$ has a solution

but not both.

How to prove it: rewrite $Ax \leq b$ as ??? (tutorial question)

Absence of arbitrage

- ▶ by Farkas' lemma, there is no arbitrage $\iff$ there exists $y \in \mathbf{R}_+^m$ with $V^T y = p$
- ▶ since first column of V is $\mathbf{1}$, we have $\mathbf{1}^T y = 1$
- ▶ y is interpreted as a risk-neutral probability on the outcomes $1, \dots, m$
- ▶ $V^T y$ are the expected values of the payoffs under the risk-neutral probability
- ▶ interpretation of $V^T y = p$: asset prices equal their expected payoff under the risk-neutral probability
- ▶ arbitrage theorem: there is no arbitrage $\iff$ there exists a risk-neutral probability distribution under which each asset price is its expected payoff

Example

$$V = \begin{bmatrix} 1.0 & 0.5 & 0.0 \\ 1.0 & 0.8 & 0.0 \\ 1.0 & 1.0 & 1.0 \\ 1.0 & 1.3 & 4.0 \end{bmatrix}, \quad p = \begin{bmatrix} 1.0 \\ 0.9 \\ 0.3 \end{bmatrix}, \quad \tilde{p} = \begin{bmatrix} 1.0 \\ 0.8 \\ 0.7 \end{bmatrix}$$

- ▶ with prices p , there is an arbitrage

$$x = \begin{bmatrix} 6.2 \\ -7.7 \\ 1.5 \end{bmatrix}, \quad p^T x = -0.2, \quad \mathbf{1}^T x = 0, \quad Vx = \begin{bmatrix} 2.35 \\ 0.04 \\ 0.00 \\ 2.19 \end{bmatrix}$$

- ▶ with prices $\tilde{p}$, there is no arbitrage, with risk-neutral probability

$$y = \begin{bmatrix} 0.36 \\ 0.27 \\ 0.26 \\ 0.11 \end{bmatrix} \quad V^T y = \begin{bmatrix} 1.0 \\ 0.8 \\ 0.7 \end{bmatrix}$$